Gene Set Enrichment Analysis using GeoMx Data

2026-05-01

SET WORKING DIRECTORY

We first set up a working directory where I intend to store all my GeoMx files that will be necessary to run QC and downstream analysis.

```
dir<-"/Users/srujansingh/Documents/Personal/GeoMx"
setwd(dir)
```

LOAD REQUIRED LIBRARIES

The following libraries need to be uploaded into the R working environment. These libraries contain the functions needed to perform downstream analysis.

```
library(clusterProfiler)
library(AnnotationDbi)
library(org.Mm.eg.db)
library(ggplot2)
library(msigdbr) #For HALLMARK Pathway
library(enrichplot) #For ClusterProfiler()
library(biomaRt)
```

LOAD THE DIFFERENTIAL EXPRESSION ANALYSIS RESULTS

Earlier, we had performed differential expression analysis to find differentually expressed genes between the normal and DKD kidneys.

```
#Loading the results from the differential expression analysis performed earlier:
results2<-readRDS("results2.rds") #results were saved as R object.
head(results2) # helps visualize the structure and elements of the result.
```

```
##                   Gene     Subset     Contrast   Estimate   Pr(>|t|)       FDR
## normal...DKD       A2M glomerulus normal - DKD  0.2006066 0.72985362 0.9168192
## normal...DKD.1   ACADM glomerulus normal - DKD  0.3045045 0.20344107 0.7171048
## normal...DKD.2   ACADS glomerulus normal - DKD -0.3855053 0.04353686 0.6322286
## normal...DKD.3   ACAT1 glomerulus normal - DKD  0.3318921 0.08130633 0.6817166
## normal...DKD.4  ACVRL1 glomerulus normal - DKD  0.2842001 0.21962478 0.7187335
## normal...DKD.5   PSEN1 glomerulus normal - DKD -0.1335426 0.32116729 0.7586819
```

SUSBET THE RESULT

Now we subset the result to focus only on the glomerulus region. Also, we only keep genes with FDR<0.05

```
res_glom<-results2[(results2$Subset=="glomerulus")&(results2$'FDR' <0.05),]
```

ARRANGE THE GENE LIST BASED ON THE ESTIMATE VALUES

Now that we have subset the differentially expressed gene list, we need to a arrange the genes as per their estimate values.

```
df_glom<-res_glom[rev(order(res_glom$Estimate)),]
head(df_glom) #visualize after rerranging
```

```
##                      Gene     Subset     Contrast  Estimate     Pr(>|t|)
## normal...DKD.7722   MAL2 glomerulus normal - DKD 1.1927189 5.027049e-04
## normal...DKD.6427 STARD7 glomerulus normal - DKD 0.7671541 4.452074e-23
## normal...DKD.653  PPP2CB glomerulus normal - DKD 0.6827211 1.960327e-24
## normal...DKD.6262   ADI1 glomerulus normal - DKD 0.6306126 1.652654e-16
## normal...DKD.5158    VCL glomerulus normal - DKD 0.5990210 1.647113e-12
## normal...DKD.902  TMEM98 glomerulus normal - DKD 0.5564434 8.136845e-12
##                            FDR
## normal...DKD.7722 4.759723e-02
## normal...DKD.6427 2.255198e-19
## normal...DKD.653  1.986007e-20
## normal...DKD.6262 5.581014e-13
## normal...DKD.5158 2.383843e-09
## normal...DKD.902  9.159375e-09
```

CREATE THE ORDERED GENE LIST FOR GENE SET ENRICHMENT ANALYSIS

Now, we need to create the gen list with their corresponding estimate values to run geneset enrichment analysis (GSEA). We are using using HALLMARK database for our geneset enrichment analysis.

```
df_glom_HALL<-df_glom$Estimate
names(df_glom_HALL)<-df_glom$Gene
head(df_glom_HALL) #visualize gene list for GSEA
```

```
##      MAL2    STARD7    PPP2CB      ADI1       VCL    TMEM98
## 1.1927189 0.7671541 0.6827211 0.6306126 0.5990210 0.5564434
```

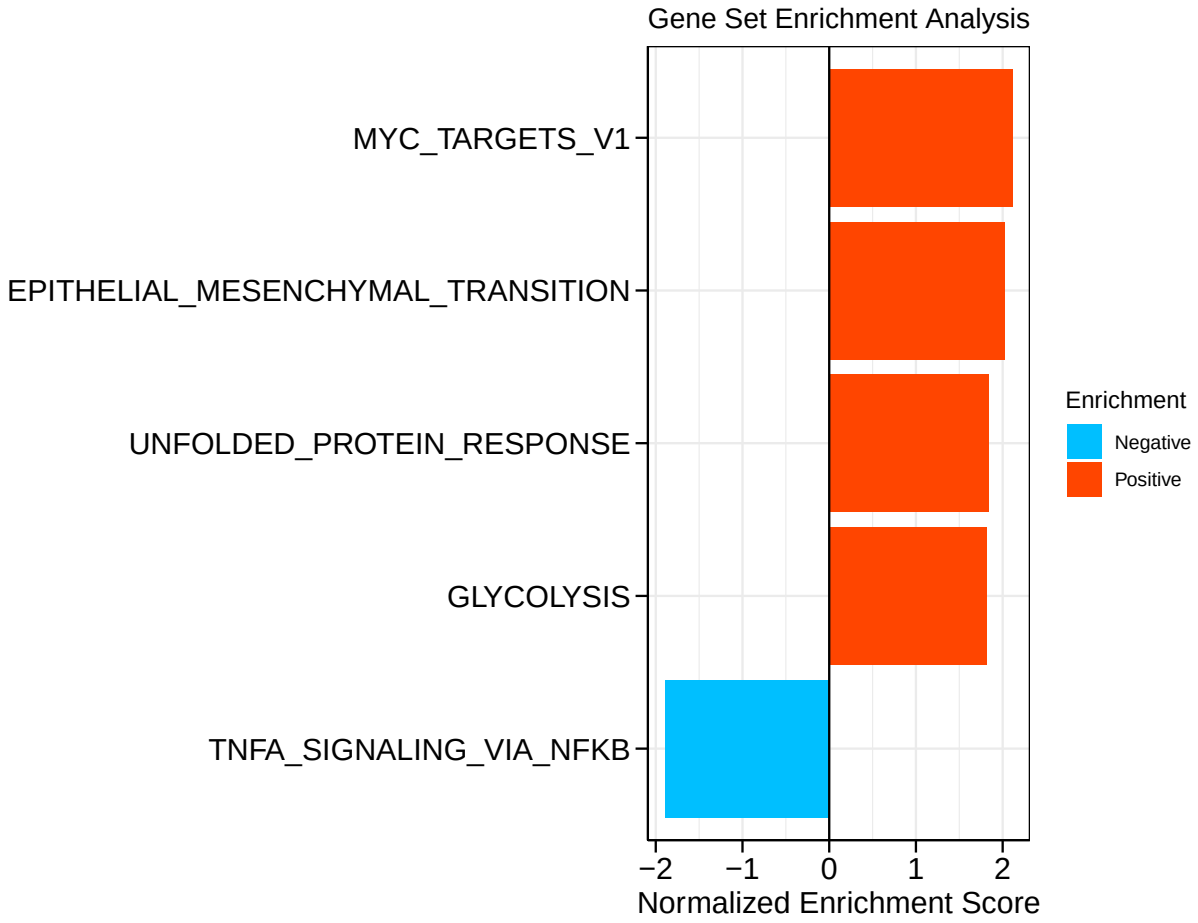
RUN THE GENE SET ENRICHMENT ANALYSIS

Since, the ordered gene list looks fine proceeed to perform gene set enrichment analysis.

```
gene_sets <- msigdbr(species = "Homo sapiens", category = "H")
background <- gene_sets[, c("gs_name", "gene_symbol")]

#df_glom.HALL.GSEA will contain the geneset enrichment analysis results
df_glom.HALL.GSEA <- GSEA(  #GSEA function
  geneList = df_glom_HALL,
  TERM2GENE = background,
  pvalueCutoff = 0.05,
  pAdjustMethod = "BH"
)
```

PROCESSING THE GENE SET ENRICHMENT DATA (FOR VISUALIZATION)

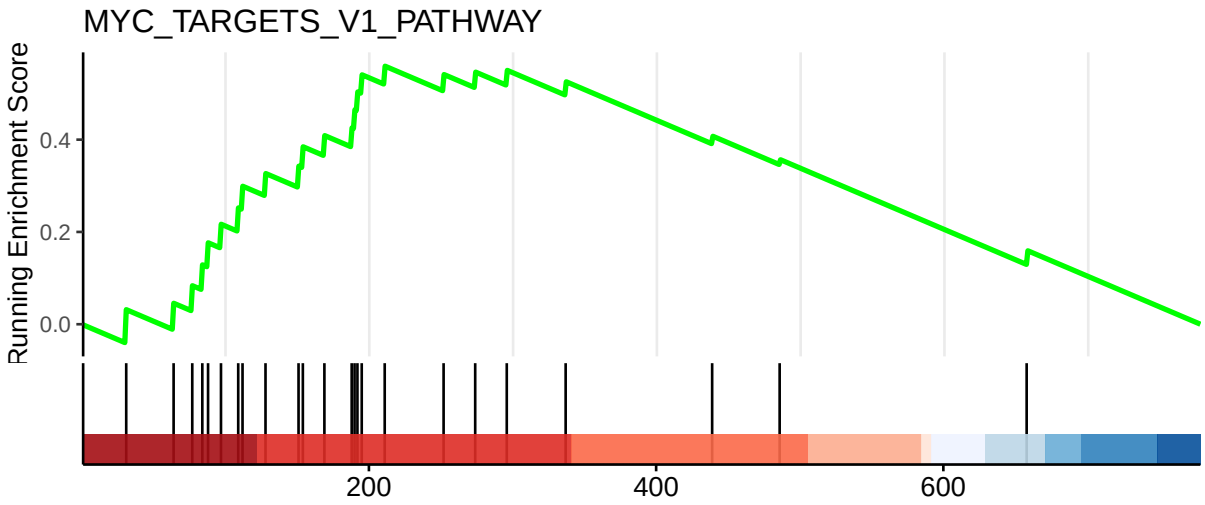
```r
# df_glom.HALL.GSEA.Plot<-df_glom.HALL.GSEA
# gseaplot(df_glom.HALL.GSEA.Plot, geneSetID = 1)
# saveRDS(df_glom.HALL.GSEA.Plot, file="EnrichmentPlots/Glomerulus_Normal_vs_DKD_HALL.GSEA.rds")
#
# if(sign(max(df_glom.HALL.GSEA$NES))==1)
# { df_glom.HALL.GSEA<-df_glom.HALL.GSEA[rev(order(df_glom.HALL.GSEA$NES)),]
# } else { df_glom.HALL.GSEA<-df_glom.HALL.GSEA[order(df_glom.HALL.GSEA$NES),] }
#
# df_glom.HALL.GSEA$core_enrichment<-gsub("/",",",df_glom.HALL.GSEA$core_enrichment)
# head(df_glom.HALL.GSEA$Description) #Visualize the enriched pathways.
#
# df_glom_vis<-data.frame(Pathway=df_glom.HALL.GSEA$Description,
#                         NES=df_glom.HALL.GSEA$NES)
#
# df_glom_vis$Pathway<-gsub("HALLMARK_","", df_glom_vis$Pathway)
# df_glom_vis$Enrichment<-"Positive"
# df_glom_vis[df_glom_vis$NES<0,]$Enrichment<-"Negative"
# head(df_glom_vis)
#
# df_glom_vis$Pathway<-factor(df_glom_vis$Pathway, levels=rev(df_glom_vis$Pathway))

# ggplot(df_glom_vis, aes(x=Pathway, y=NES, fill=Enrichment)) +
#   geom_bar(stat="identity") +
#   coord_flip() +
#   geom_hline(yintercept=0, linetype="solid", color="black", size=0.5) +
#   coord_flip() + theme_minimal()+
#   scale_fill_manual(values = c("Positive" = "orangered", "Negative" = "deepskyblue")) +
#   labs(x = "", y = "Normalized Enrichment Score", title = "Gene Set Enrichment Analysis") +
#   theme(panel.border = element_rect(color = "black", fill = NA, size = 0.5),  # Add box around plot
#         axis.text.x = element_text(size = 14, color="black"), # Increase font size of pathway names
#         axis.text.y = element_text(size = 14, color="black"),
#         axis.ticks = element_line(color = "black"),  # Add tick marks
#         axis.ticks.length = unit(0.2, "cm"),  # Adjust tick mark length
#         axis.line = element_line(color = "black"),
#         axis.title.x = element_text(size = 14))
# dev.off()

#The above code is commented out becasue it was giving error ithin the Rmd file.
#But if you run it within the R console, you should get a figure as shown below:
knitr::include_graphics("NES_GeoMx_Glom.pdf")
```



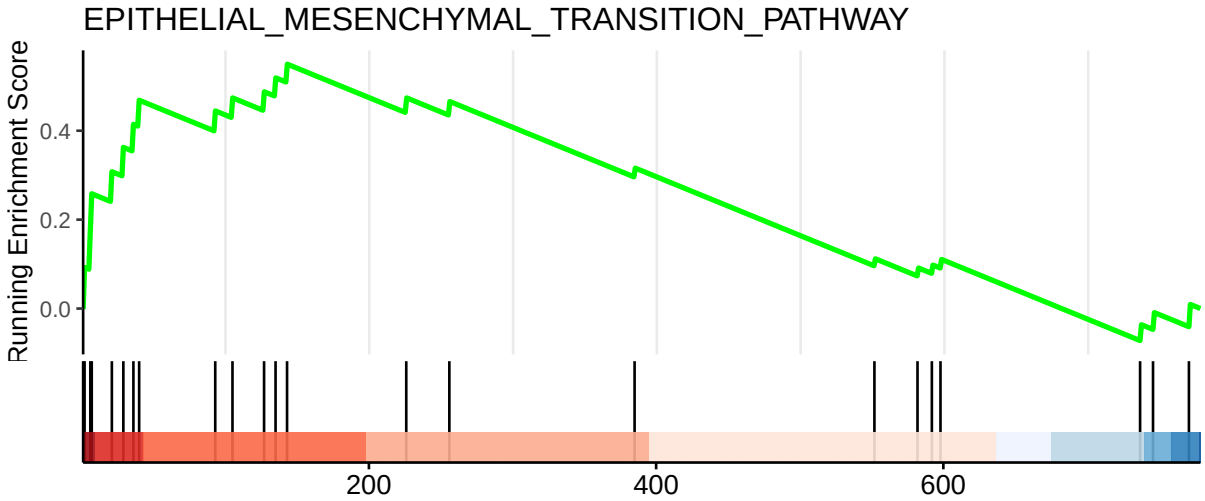
You can visualize all the enriched pathways along with their normalized enrichment scores (NES). A positive enrichment (NES) is colored red while a negative enrichment (NES) is colored blue.

#VISUALIZE INDIVIDUAL RUNNING ENRIECHMENT SCORE PLOTS FOR INDIVIDUAL PATHWAYS

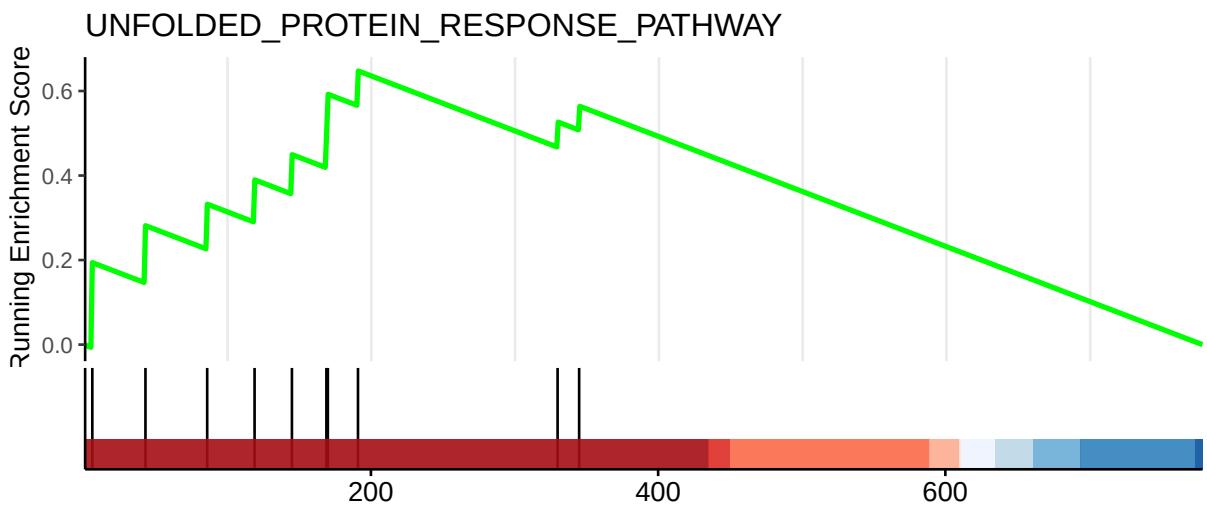
```
# ID1="HALLMARK_MYC_TARGETS_V1"
# gseaplot2(df_glom.HALL.GSEA.Plot, geneSetID = c(ID1),  subplots=1:2, title="MYC_TARGETS_V1_PATHWAY")

#Similarly, the above code has been commented out. But if you run it in the R console
#you should see the follwoing plot:
knitr::include_graphics("MYC.pdf")
```



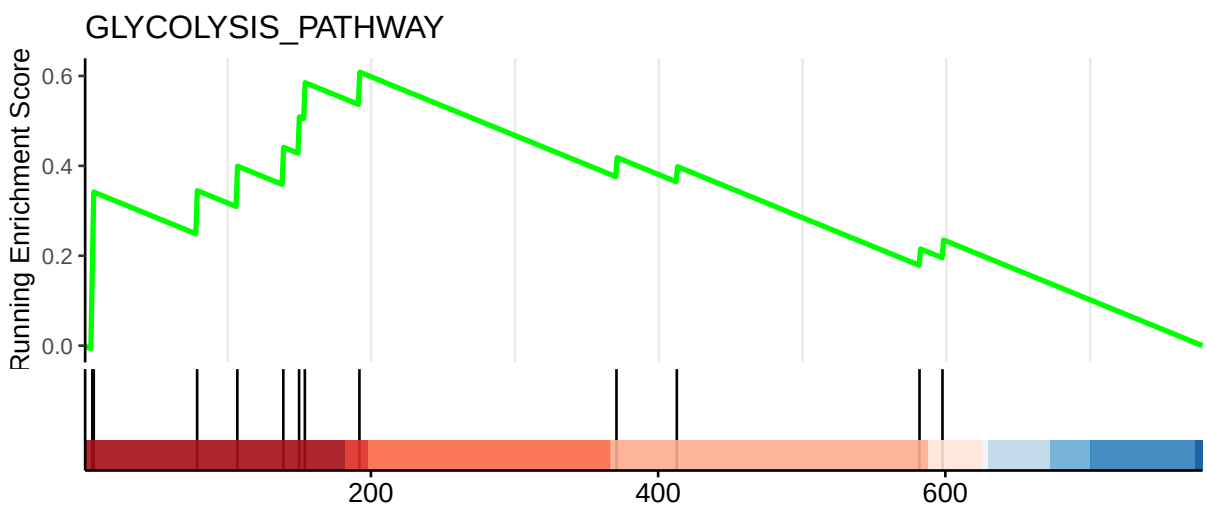

```
# ID2="HALLMARK_EPITHELIAL_MESENCHYMAL_TRANSITION"
# gseaplot2(df_glom.HALL.GSEA.Plot, geneSetID = c(ID2),  subplots=1:2, title="EPITHELIAL_MESENCHYMAL_TR
#
# ID3="HALLMARK_UNFOLDED_PROTEIN_RESPONSE"
# gseaplot2(df_glom.HALL.GSEA.Plot, geneSetID = c(ID3),  subplots=1:2, title="UNFOLDED_PROTEIN_RESPONSE
#
# ID4="HALLMARK_GLYCOLYSIS"
# gseaplot2(df_glom.HALL.GSEA.Plot, geneSetID = c(ID4),  subplots=1:2, title="GLYCOLYSIS_PATHWAY")
#
# ID5="HALLMARK_TNFA_SIGNALING_VIA_NFKB"
# gseaplot2(df_glom.HALL.GSEA.Plot, geneSetID = c(ID5),  subplots=1:2, title="TNFA_SIGNALING_VIA_NFKB_P

#Similarly, the above code has been commented out. But if you run it in the R console
#you should see the follwoing plot:
knitr::include_graphics("EMT.pdf")
```



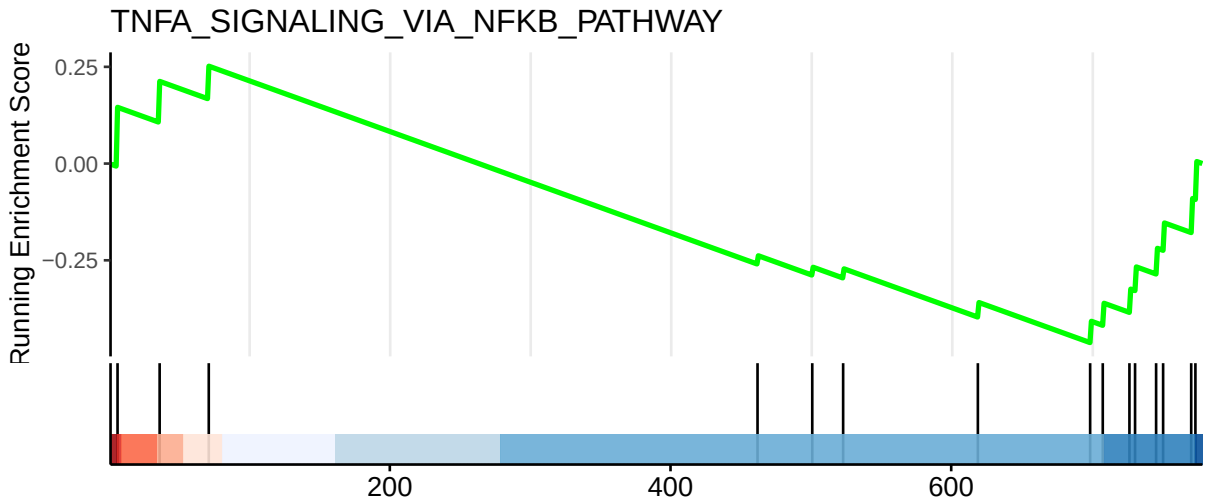
```r
knitr::include_graphics("Unfolded_Protein.pdf")
```




```r
knitr::include_graphics("Glycolysis.pdf")
```

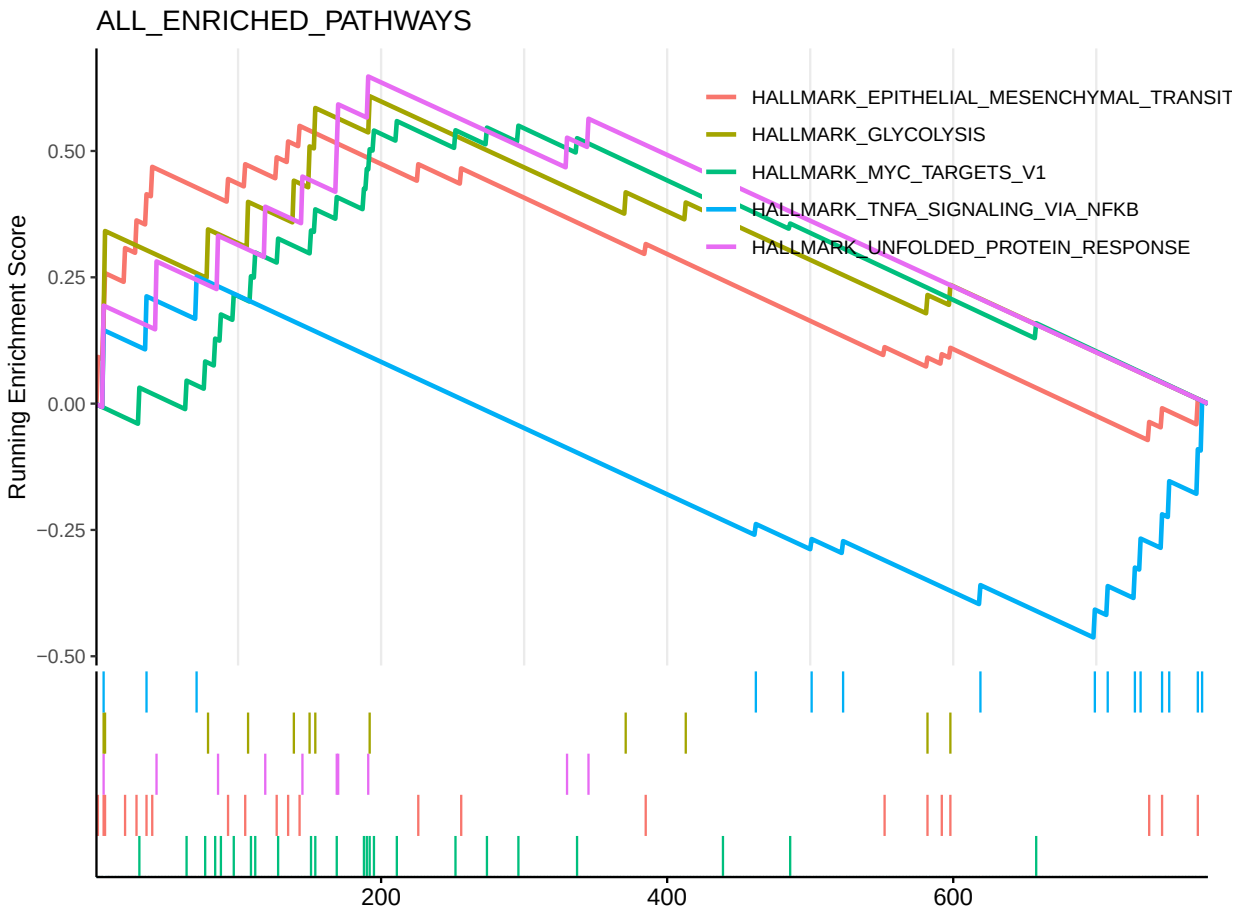



```r
knitr::include_graphics("NFKB.pdf")
```



Finally, you can also visualize all the running score enrichment plots within the same plot:

```
#gseaplot2(df_glom.HALL.GSEA.Plot, geneSetID = c(ID1, ID2, ID3, ID4, ID5),  subplots=1:2, title="ALL_EN

#Similarly, the above code has been commented out. But if you run it in the R console
#you should see the follwoing plot:
knitr::include_graphics("All_Pathways.pdf")
```


ALL_ENRICHED_PATHWAYS
HALLMARK_EPITHELIAL_MESENCHYMAL_TRANSIT
HALLMARK_GLYCOLYSIS
HALLMARK_MYC_TARGETS_V1
HALLMARK_TNFA_SIGNALING_VIA_NFKB
HALLMARK_UNFOLDED_PROTEIN_RESPONSE
Running Enrichment Score
0.50
0.25
0.00
−0.25
−0.50
200
400
600